

Final Report

Auto-Ticket Classification using Flow Designer

Comprehensive Project Report

1. Abstract

This project implements an automatic ticket classification system for a school IT helpdesk using ServiceNow Flow Designer. Students submit support requests through a custom School IT Ticket form, and a Flow Designer workflow automatically classifies each ticket by detecting keywords in the short description. The solution eliminates manual category selection, reduces triage workload, and provides immediate email acknowledgment to the requester.

2. Introduction

School IT helpdesk teams handle a variety of student requests ranging from Wi-Fi issues to hardware faults and password resets. Without automation, technicians must manually review and categorize every incoming ticket, a repetitive process that delays resolution. This project addresses that challenge by implementing a keyword-based classification flow within ServiceNow.

3. Objectives

- Simplify ticket submission by removing manual category selection.
- Automate classification using keyword detection in Flow Designer.
- Provide immediate email acknowledgment to students.
- Deliver pre-classified tickets to technicians for faster resolution.
- Track all customizations through ServiceNow Update Sets.

4. Literature Survey

#	Topic	Key Insight
1	ServiceNow Flow Designer	Event-driven, no-code automation for record-based workflows.
2	IT Service Management (ITSM)	Structured ticket handling improves response time and accuracy.
3	Keyword-Based Classification	Rule-based systems are effective for well-defined issue domains.
4	Email Notification Systems	Immediate acknowledgment builds trust and transparency.

5. Methodology

The project follows an Agile methodology with 5 sprints, each mapped to a project milestone:

Sprint	Milestone	Key Deliverables
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1	Requirement Analysis	Problem definition, Update Set creation
2	Backend Development	Custom table, fields, dependent choices
3	Flow Designer Automation	Classification flow, email notification
4	Testing & Validation	Functional testing, UAT
5	Deployment & Packaging	Update Set completion, XML export

6. Implementation

6.1 Custom Data Model

A custom School IT Ticket table was created with fields: Number (auto-generated), Caller (reference), Short Description (string), Category (choice), Subcategory (dependent choice), State, and Assignment Group.

6.2 Flow Designer Workflow

The Auto Classify School IT Tickets flow triggers on Record Created. It evaluates the Short Description against keyword conditions and sets the appropriate Category and Subcategory values. An email action sends an acknowledgment to the Caller.

6.3 Classification Rules

Keywords	Category	Subcategory
Wi-Fi, Network, Connectivity	Network	Wi-Fi / Connectivity
Projector, Hardware, Display	Hardware	Projector / Display
Password, Login, Access	Access	Password / Login
Slow, Performance, Lag	Performance	System Performance

6.4 Security & Change Tracking

ACLs and roles control access to the School IT Ticket table. All customizations (table definitions, field configurations, choice values and flow logic) are captured in the Project Update Set and exportable as XML for version control and deployment.

7. Results

All functional test cases and UAT scenarios passed with 100% acceptance rate.

Test Area	Cases	Passed	Failed
Keyword Classification	4	4	0
Email Notification	1	1	0
No-Match Handling	1	1	0
UAT Scenarios	5	5	0

8. Advantages & Disadvantages

Advantages

- Simple requester experience with basic input fields.
- Automatic classification for supported issue keywords.
- Consistent Category/Subcategory mapping.
- Immediate requester acknowledgment via email.
- Controlled change tracking through Update Sets.

Disadvantages

- Keyword rules depend on how the problem is described.
- Unsupported or vague descriptions require manual review.
- Current prototype is rule-based, not model-based.
- No automatic technician assignment implemented.

9. Conclusion

The Auto-Ticket Classification using Flow Designer project demonstrates a complete ServiceNow workflow for simplifying school IT helpdesk ticket processing. The implementation combines a custom data model, dependent choices, event-driven classification, email acknowledgment, access control, and Update Set based deployment tracking. All test cases passed, validating the solution's effectiveness.

10. Future Scope

- Expand keyword coverage and add fallback handling for ambiguous requests.
- Introduce automatic assignment to support groups.
- Add status-change and resolution notifications.
- Build dashboards for ticket volume, classification, and response trends.
- Evaluate AI-assisted classification as a future enhancement.

11. Appendix

Source Code: No separate application code - implemented via ServiceNow configuration.

Dataset: Not applicable - rule-based classifier, no ML dataset required.

GitHub: <https://github.com/Akash717656/Auto-Ticket-Classification-using-Flow-Designer>